

UMAIR ANWAR

Computer Science Undergraduate | Backend Engineer

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ABOUT ME

A computer science undergraduate at GIKI with a strong interest in backend engineering and building scalable systems. I have experience developing full-stack applications using C++, TypeScript, Node.js, Express.js, and PostgreSQL. Currently learning Java and Spring Boot, with a focus on system design, high-traffic backend systems, automation, distributed systems, and tools such as message queues, and caching.

SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, SQL, Python, x86-64 and RISC-V Assembly, Bash

Frameworks & Databases: React.js, Node.js, Express.js, PostgreSQL, MySQL, MongoDB

Authentication & APIs: JWT, OAuth 2.0, REST APIs

Automation & Tools: Git/GitHub, n8n, Docker, Postman, GDB, Cron Jobs

Core Concepts: Web Sockets, Data Structures & Algorithms, Object-Oriented Programming, Database Design

EXPERIENCE

Contour Software

Database Administration & Software Development Intern

June 2026 – July 2026

- Completed a six-week rotational internship across Database Administration and Software Development, gaining hands-on exposure to user roles and permissions, backups, triggers, logs, and database indexing.
- Designed and developed a full-stack web application using HTML, CSS, TypeScript, Vite, Node.js, Express.js, and PostgreSQL; designed the database schema, built REST API, integrated the frontend and backend, and implemented JWT and OAuth 2.0 authentication.
- Deployed the production frontend on Vercel and backend on Render, while working within a professional Git workflow.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Bachelor of Computer Science

2024 – 2028 (Expected)

PROJECTS

MDCATemy

React.js, Node.js, Express.js, PostgreSQL

Developed a scalable web application for an MDCAT preparation academy, enabling students to make custom quizzes, attempt tests, bookmark questions, and track their performance through detailed analytics. Built admin features for payment verification, MCQ bank management through Excel uploads, and custom test creation for selected students.

Roamistan

HTML/CSS, TypeScript, Express.js, Node.js, PostgreSQL

Roamistan is a virtual tourism platform that lets people explore tourist places across Pakistan before visiting them. Users can walk through locations in immersive 3D and 360 views, bookmark their favorite places, and learn about their history. The project also helps hotels, museums, travel agencies, and other tourism businesses promote their destinations through interactive virtual experiences.

LiveNums

TypeScript, React.js, Express.js, Node.js, Web Sockets

Built a real-time web app using Node.js, Express, WebSockets, TypeScript, and React, with instant server-to-client state updates without polling. Added cron scripts to generate automatic state changes every second, along with JWT authentication and username-based rate limiting for WebSocket connections and messages.

ClearCash

TypeScript, React.js, Express.js, Node.js, PostgreSQL

ClearCash is a web-based e-wallet that solves the problem of spending, organizing money, and tracking expenses in a single app. ClearCash removes the need for a separate budgeting app by providing virtual spending pockets, budget tracking, fund transfers, notifications, monthly transaction history, and a built-in calculator in a single app to help you organize and manage personal finances effortlessly.

PhotoDukaan

C++, Qt, Image Processing

Built a desktop image editor in C++ from scratch, implementing raw pixel manipulation algorithms for image processing, color grading, blur effects, and custom filters, including edge detection, heat maps, and other visual transformations without relying on prebuilt image-processing libraries.

ACHIEVEMENTS

- Maximum Codeforces Rating: 1113
- Merit Scholarship Holder at GIKI
- ICPC Regional Finalist
- STFS Scholarship Holder
- Four-time Dean's Honor List recipient
- Best Presentation Award at Contour Software